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Distributed AI: Radar explanation

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Introduction

CARLA's simulated RADAR was used to estimate distances to the lead vehicle. The choice between RADAR, LiDAR and depth cameras was based off their robustness and efficacy in real-life applications. Specifically, for distance estimation we looked at range, distance accuracy and its robustness to weather conditions. Based on [this study](#), we determined that RADAR was the most suitable candidate.

RADAR – Working

Callback: Processes radar data for every tick. This radar data contains all detections for that tick. Each detection is characterized by altitude (β), azimuth (α), distance (d) and velocity. Here we perform a mapping of a **radar detection** to a **world point**.

(1.1) Build a forward vector along the radar's local X by the measured range d .

$$f = [d - 0.25, 0, 0]$$

(1.2) Rotate this vector by the detection's α and β around the radar, expressed in radar's current world orientation.

$$f' = R(\text{sensor yaw} + \alpha, \text{sensor pitch} + \beta, \text{sensor roll}) f$$

(1.3) Translate into world by adding the radar's world location t_{wr} .

$$X_w = t_{wr} + f' \rightarrow (x_w, y_w, z_w)$$

This provides us with the true 3D point where the radar saw the return. This is stored to a shared memory alongside the velocity and distance.

Camera intrinsics: using the pinhole matrix K , we convert camera-frame meters into pixels.

(2.1) Pixel focal lengths based on field of view (FOV) and frame width (w) and height (h).

$$f_x = \frac{w}{2 \tan(\frac{HFOV}{2})} \quad f_y = \frac{h}{2 \tan(\frac{VFOV}{2})}$$

(2.2) Pinhole matrix K , where c_x and c_y represent the image center.

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

Camera extrinsics: The CARLA world and sensors are expressed in Unreal axes: X forward, Y right, Z up. A standard CV projection requires camera CV axes: x right, y down, z forward. We create a transform P , that projects world coordinates to CV camera coordinates.

(3.1) The inverse matrix of the camera transform T_{wc}^{ue} is fetched from the camera sensor in CARLA and is used to project UE world coordinates to UE camera coordinates. It is also used in calculating P . $R_{ue \rightarrow cv}$ performs a constant basis-change of the axes.

$$P = \begin{bmatrix} R_{ue \rightarrow cv} & 0 \\ 0 & 1 \end{bmatrix} T_{wc}^{ue} \quad \text{with} \quad R_{ue \rightarrow cv} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

Projection: We use the previously calculated matrices K and P to perform the following mapping:

$$\text{World (UE)} \rightarrow \text{Camera (UE)} \rightarrow \text{Camera (CV)}$$

(4.1) Convert each world point to its homogeneous version.

$$X_{w,i} = [x_w, y_w, z_w]^T \rightarrow X_{w,i,h} = [x_w, y_w, z_w, 1]^T$$

(4.2) Transform world points to CV camera frame

$$X_{w,i,h} P = [x_c, y_c, z_c, 1]^T$$

(4.3) Apply intrinsics (pinhole model), mapping camera points to pixels

$$[x_c, y_c, z_c]^T K = [f_x x_c + c_x z_c, f_y y_c + c_y z_c, z_c]^T$$

(4.4) Then we can extract the horizontal pixel u and vertical pixel v as such

$$u = f_x \frac{x_c}{z_c} + c_x \quad \text{and} \quad v = f_y \frac{y_c}{z_c} + c_y$$

Provided that object detection has produced a bounding box of the lead vehicle, and that this bounding box is within the lane of the ego vehicle, we can then determine the distance to the lead vehicle by looking at the collection of radar points with horizontal and vertical pixel values that are located within that bounding box.

Lastly, one distance is filtered out from this collection of radar points. For safety reasons (more conservative driving), we consider the point with minimal distance out of the collection.

A limitation to this approach is that bounding boxes that do not contour the car perfectly, may encapsulate parts of the ground that are closer to the ego vehicle than the lead vehicle. However, the difference in distance is relatively small in this case. Another limitation occurs when other objects appear in between the lead vehicle and the ego vehicle, in this case, the distance of that object would be considered as it is closer to the ego vehicle. Although a real-life scenario where this may occur would be rare, an approach to fix this would be to cluster the distances of the RADAR points and take the minimum value of the largest cluster, assuming that the object in between the ego vehicle and the lead vehicle is small enough that it does not prevent the detection of the lead vehicle in the first place.